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# Total Cloud Cover Forecasting for EO Tasking

*ML Weather Models from Keisler-2022 to GenCast, with a Differentiable-Pipeline Lens*

LATEX → PDF · ~40–50 PAGES

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## § Abstract


Cloud cover is not a normal weather variable. It is bounded $[0, 1]$, strongly bimodal (lots of 0s and 1s), spatially intermittent, and — critically — a *diagnostic* in IFS rather than a prognostic state. ERA5's `tcc` is itself a model output derived from cloud water/ice plus saturation, so any model trained on it learns to mimic the IFS parameterization rather than ground truth. Naïve MSE collapses toward 50% gray mush. The right architectural choices — logit transform or beta head, CSI/Brier/HSS validation at operational thresholds, calibration against geostationary truth — fall out of taking that constraint seriously.

This monograph surveys the ML-weather landscape as of early 2026, with a focus on what is actually useful for Earth-observation collection tasking on constrained hardware.

CONTENTS

01 Introduction and the cloud-cover-is-not-weather framing
02 Where the field actually is: Keisler, GraphCast, AIFS, GenCast, Aurora, NeuralGCM
03 Probabilistic vs deterministic for EO tasking
04 Hardware and Compute Tiers (Tier 1 consumer through Tier 4 research clusters)
05 Operational stack: multi-source fusion by lead time
06 Calibration against observation truth vs reanalysis
07 Local inference vs operational center inference
08 The ∂LITE pattern: differentiable orbit ∘ geometry ∘ atmosphere
09 TCC nowcasting: an open opportunity
10 Notebook walk-through: ECMWF Open Data → in-between interpolation → ARCO ERA5 validation
11 Appendix A: Variable-name reference across GRIB2 / ERA5 / ECMWF
12 Appendix D: Skill metrics formulae (RMSE, MAE, Brier, CSI, HSS)

## § Where the field actually is

The honest landscape, with model · year · output · resolution · cadence · TCC skill · open weights:

- **Keisler-2022** · 2022 · Deterministic · 1° · 6h · demonstrated, weak · yes
- **GraphCast** · 2023 · Deterministic · 0.25° · 6h · decent · yes
- **Pangu-Weather** · 2023 · Deterministic · 0.25° · 1/3/6/24h · decent · yes
- **AIFS (ECMWF)** · 2024 · Deterministic + ENS · 0.25° · 6h · *best in class* · yes
- **GenCast** · 2024 · Probabilistic (diffusion ensemble) · 0.25° · 12h · strong, uncertainty-aware · yes
- **Aurora** · 2024 · Foundation, fine-tunable · 0.25° · 6h · strong · yes

The MAUSAM paper (arXiv 2509.01879, Sep 2025) benchmarks all of these against ground-based and geostationary observations — not just ERA5 — and finds AIFS most consistent overall, with GraphCast and GenCast close behind. All beat Keisler-2022 substantially. Errors against actual observations are 15–45% larger than errors against reanalysis, which is the relevant gap for an EO tasking application.

## § Hardware tiers

**Tier 1 (consumer, 8–16 GB VRAM)** — RTX 5060 Ti (16 GB), 4060 Ti (16 GB). Comfortable: Keisler-2022, GraphCast-small (1°), FourCastNet v1, Pangu (no TCC), NeuralGCM (1.4°/2.8°). Tight: AIFS-single (0.25°, ~13 GB with flash-attn), GraphCast (0.25°), GenCast (1°). Won't fit: GenCast (0.25°, needs 60 GB), Aurora MR/HR.

**Tier 2 (prosumer, 24–48 GB)** — A6000, 4090, 5090. Adds AIFS-single comfortably, GraphCast at fp32, Aurora MR.

**Tier 3 (enterprise, 80 GB+)** — A100/H100. GenCast 0.25° (~25 min/member), Aurora HR (0.1°), training-from-scratch budgets.

## § The ∂LITE pattern

The differentiability of neural forecasters enables a class of inverse-problem applications (sensitivity analysis, ML-native data assimilation, gradient-based tasking optimization) that physics models cannot match. For SSA and EO applications specifically, the ∂LITE pattern is the most promising design space: compose differentiable orbit, geometry, and atmospheric operators; optimize end-to-end for collection utility.

The proprietary IP is the calibration head and the historical (forecast, observed-at-collect) archive, not the forecaster itself.

## § TCC nowcasting: an open opportunity

Most nowcasting work targets precipitation. For TCC nowcasting (0–6 h cloud cover from geostationary imagery), practical options reduce to: (1) optical-flow extrapolation of geostationary cloud mask (classic, robust, ~1 h skillful horizon); (2) custom small ML model trained on (GOES L1B sequence) → (GOES L2 cloud mask future); (3) persistence + climatology blend at the lowest-effort end. *This is a real opportunity for proprietary EO-tasking IP: there is no strong public TCC nowcaster as of early 2026.*

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## § Status & provenance

Produced in conversation b47b9a50 on 2026-05-07. Accompanied by a Colab notebook (`cloud_cover_forecasting.ipynb`, 26 cells) that exercises the full ECMWF Open Data → temporal interpolation → ARCO ERA5 validation pipeline.